TECHNICAL SUPPLEMENT

2028 Olympic Games Optimization

Prepared for SportsFanatics

This appendix documents the data and methods behind every figure in our presentation.

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Each section is a self-contained reference. Appendices A–F give the full tables behind the figures shown on screen; the Methodology Summary states data sources, models, and assumptions for all three stakeholders.

Methodology Summary

Host City — Los Angeles (Appendix A)

Source: historical Olympic athlete-event records (1896–2016) joined to NOC region data. Reference Games: **Rio 2016 (Summer)**, the most recent complete Summer Games in the data. Each sport is mapped to its official LA28 venue zone (March 2026 venue plan); multi-venue sports are assigned to their primary sub-venue. We count **unique athletes** (by athlete ID) per zone, so no athlete is double-counted. Scope: the **31 returning LA28 sports**, recorded as 34 categories in the dataset (Rhythmic Gymnastics & Trampolining group under Gymnastics; Beach Volleyball under Volleyball). Total athletes analysed: **11,179**.

National Olympic Committees (Appendix B)

We add World Bank external data — **GDP per capita (NY.GDP.PCAP.CD)** and **population (SP.POP.TOTL)**, 2016 — and model Rio 2016 medal counts with a **Poisson regression** on log(population) and log(GDP per capita). Poisson is the correct family because medals are non-negative count data. Expected medals are estimated **out-of-fold** (5-fold CV, no leakage); the residual (actual − expected) is the Resource-Adjusted Performance Index (RAPI). Out-of-fold **R^2 = 0.80** over 182 countries. Each underperformer is matched to its nearest **overperforming** peer by Euclidean distance in log-population / log-GDP space.

Athletes (Appendices C & D)

Strategy clustering: KMeans (k=2) groups countries as specialist vs generalist by how concentrated their all-time Summer medals are across sports, using event-level medal counts and a concentration feature set (Herfindahl index + effective number of sports); countries with fewer than 20 medals are excluded as statistically unstable. **Peak age windows:** for each sport with at least 100 medal-winning athletes on record, we report the median medal-winning age and the interquartile range (middle 50%).

Consistency note. All medal-based figures use Rio 2016 as the reference Games and event-level medal counts (one medal per event, not per athlete) unless a section states an all-time scope. Every number in this document is computed directly from the datasets and notebooks in our repository.

Appendix A — Host-City Zone Load

Unique athletes per LA28 venue zone, Rio 2016 reference. Top three zones (Downtown/Exposition Park, Long Beach, Carson) carry **77%** of all 11,179 athletes.

Zone	Athletes	Share
Downtown/Exposition Park	4,581	41.0%
Long Beach	2,455	22.0%
Carson	1,526	13.7%
Inglewood	1,223	10.9%
Pasadena	609	5.4%
Anaheim	283	2.5%
Arcadia	200	1.8%
Pacific Palisades	120	1.1%
Venice	110	1.0%
San Fernando Valley	72	0.6%

Appendix A — Sport-to-Zone Mapping

Every sport mapped to its LA28 venue zone, with the number of Rio 2016 athletes in that sport.

Sport	Zone	Athletes
Volleyball	Anaheim	283
Equestrianism	Arcadia	200
Cycling	Carson	513
Hockey	Carson	390
Rugby Sevens	Carson	299
Tennis	Carson	196
Archery	Carson	128
Athletics	Downtown/Expo	2,269
Judo	Downtown/Expo	389
Wrestling	Downtown/Expo	346
Boxing	Downtown/Expo	283
Weightlifting	Downtown/Expo	255
Fencing	Downtown/Expo	245
Gymnastics	Downtown/Expo	196
Table Tennis	Downtown/Expo	172
Badminton	Downtown/Expo	172
Taekwondo	Downtown/Expo	126
Rhythmic Gymnastics	Downtown/Expo	96
Trampolining	Downtown/Expo	32
Swimming	Inglewood	942
Basketball	Inglewood	281
Rowing	Long Beach	546
Shooting	Long Beach	390
Sailing	Long Beach	380
Handball	Long Beach	353
Canoeing	Long Beach	330
Water Polo	Long Beach	258
Synchronized Swimming	Long Beach	102
Beach Volleyball	Long Beach	96
Golf	Pacific Palisades	120
Football	Pasadena	473
Diving	Pasadena	136
Modern Pentathlon	San Fernando Valley	72
Triathlon	Venice	110

Appendix B — NOC Over / Underperformers

Countries ranked by how far their Rio 2016 medal haul sits above or below the resource-adjusted baseline — a model of expected medals given a country's population and economic capacity. Overperformers beat their baseline; underperformers fall short. Each list runs from the largest gap to the smallest.

B.1 — Overperformers (all 44 countries)

Country	Country	Country
Russia	Belarus	Canada
UK	Serbia	Bahamas
Azerbaijan	Ethiopia	Greece
Kazakhstan	Georgia	Bulgaria
New Zealand	Germany	Burundi
Hungary	Czech Republic	Grenada
Kenya	Armenia	Slovakia
Uzbekistan	Netherlands	Tajikistan
Jamaica	Lithuania	Fiji
Australia	Italy	Niger
Ukraine	Colombia	Kosovo
France	Slovenia	China
Cuba	Poland	Ivory Coast
Croatia	Tunisia	Bahrain
Denmark	Mongolia	

B.2 — Underperformers (all 66 countries)

Country	Country	Country
India	Iraq	Iceland
USA	Qatar	Paraguay
Saudi Arabia	Brazil	Tanzania
Japan	Luxembourg	Dominican Republic
Mexico	Egypt	Sudan

Country	Country	Country
Austria	Oman	Latvia
Indonesia	Ecuador	Cameroon
United Arab Emirates	Vietnam	Boliva
Switzerland	Sri Lanka	Democratic Republic of the Congo
Ireland	Costa Rica	Malaysia
Israel	Panama	El Salvador
Finland	Algeria	Cambodia
Norway	Uruguay	Uganda
Chile	Guatemala	South Korea
Argentina	Lebanon	Romania
Nigeria	Angola	Thailand
Belgium	Libya	Sweden
Bangladesh	Ghana	Jordan
Philippines	Cyprus	Spain
Peru	Morocco	Iran
Portugal	Myanmar	Estonia
Pakistan	Turkmenistan	Venezuela

B.3 — Peer Benchmark: recommended peer

For each underperformer, the overperforming country with the most similar resource profile (population and economic capacity) — the benchmark whose model that NOC should study and adapt.

Country	Recommended peer
Jordan	Belarus
Libya	Bulgaria
Portugal	Czech Republic
Lebanon	Bulgaria
Turkmenistan	Serbia
Ghana	Ivory Coast
Paraguay	Serbia
Dominican Republic	Cuba
Cyprus	Bahrain
Argentina	Poland
Panama	Croatia
Angola	Ivory Coast
South Korea	Italy
Morocco	Uzbekistan
Finland	New Zealand
Mexico	Russia
Costa Rica	Croatia
Uruguay	Lithuania
Oman	Slovakia
Boliva	Tunisia
Romania	Kazakhstan
Myanmar	Kenya
Ireland	Denmark

Country	Recommended peer
Iraq	Colombia
Latvia	Lithuania
Thailand	Colombia
Democratic Republic of the Congo	Ethiopia
Belgium	Netherlands
Peru	Colombia
Sudan	Kenya
Japan	Germany
El Salvador	Serbia
Austria	Denmark
Tanzania	Kenya
Cambodia	Ivory Coast
United Arab Emirates	Denmark
Israel	Denmark
Sweden	Netherlands
Switzerland	Denmark
Sri Lanka	Uzbekistan
Iran	Colombia
Chile	Greece
Saudi Arabia	Poland
Iceland	Bahamas
Uganda	Niger
Vietnam	Ukraine

Country	Recommended peer
Estonia	Bahrain
Ecuador	Kazakhstan
Cameroon	Ivory Coast
Norway	Denmark
Venezuela	Uzbekistan
Spain	Italy
Malaysia	Poland
Algeria	Colombia
Brazil	Russia
Guatemala	Tunisia

Country	Recommended peer
Qatar	New Zealand
Egypt	Ukraine
Philippines	Ukraine
Bangladesh	Ethiopia
Pakistan	Ethiopia
Indonesia	Russia
Luxembourg	Bahamas
Nigeria	Ethiopia
USA	Germany
India	China

Appendix C — Country Strategy Classification

KMeans (k=2) on all-time Summer event-level medals, classifying countries as **Specialist** (medals concentrated in 1–2 sports) or **Generalist** (spread across many). 62 countries meet the ≥ 20-medal threshold (22 specialist, 40 generalist). **Top sport** is the single discipline in which a country has won the most medals.

Specialists (most concentrated first)

Country	Medals	Top sport
Ethiopia	53	Athletics
Jamaica	76	Athletics
Kenya	100	Athletics
Morocco	23	Athletics
Trinidad	21	Athletics
Indonesia	30	Badminton
Iran	68	Wrestling
Turkey	95	Wrestling
Slovakia	28	Canoeing
Georgia	32	Wrestling
Thailand	30	Boxing

Country	Medals	Top sport
Nigeria	24	Athletics
Azerbaijan	44	Wrestling
Uzbekistan	33	Boxing
Ireland	33	Boxing
Mongolia	26	Wrestling
Taiwan	24	Taekwondo
Egypt	27	Weightlifting
Portugal	24	Athletics
Finland	310	Athletics
India	28	Hockey
Kazakhstan	70	Boxing

Generalists (most diversified first)

Country	Medals	Top sport
South Africa	88	Athletics
Estonia	34	Wrestling
Australia	518	Swimming
Cuba	214	Boxing
North Korea	54	Weightlifting
Slovenia	23	Judo
USA	2544	Athletics
Bulgaria	217	Wrestling
Colombia	28	Cycling
Argentina	74	Boxing
Japan	437	Gymnastics
New Zealand	118	Athletics
Mexico	67	Diving
Norway	155	Shooting
Romania	303	Gymnastics
Switzerland	209	Gymnastics
Lithuania	25	Athletics
Greece	151	Athletics
Belarus	84	Athletics
Hungary	504	Fencing

Country	Medals	Top sport
Netherlands	297	Swimming
Canada	305	Athletics
Czech Republic	207	Athletics
Brazil	128	Judo
Poland	290	Athletics
Sweden	513	Athletics
UK	892	Athletics
Croatia	33	Athletics
South Korea	263	Judo
Russia	1577	Athletics
Italy	608	Fencing
Germany	1389	Athletics
Serbia	107	Wrestling
Belgium	166	Cycling
Austria	108	Canoeing
Ukraine	126	Athletics
Denmark	210	Sailing
China	545	Diving
France	770	Fencing
Spain	148	Sailing

Footnote — low-medal countries. Countries with fewer than 20 event-level medals are excluded because share-based concentration is unstable at low counts. By inspection these are typically clear specialists: e.g. **Pakistan** (10 medals, 80% in Hockey) is intuitively a specialist but falls below threshold, so it is reported here rather than given an unstable cluster label. Labels reflect degree of concentration, not which sport — a country specialised in an unusual sport (e.g. India in field hockey) is correctly a specialist.

Appendix D — Peak Age Windows per Sport

Median medal-winning age and interquartile range (middle 50%) for every Summer sport with at least **100** medal-winning athletes on record, sorted youngest to oldest. Q1 / Q3 are the 25th / 75th age percentiles; IQR = Q3 − Q1. Discontinued events (Art Competitions, 15-a-side Rugby) are excluded.

Sport	Median	Q1	Q3	IQR
Rhythmic Gymnastics	18	17	20	3.2
Swimming	21	18	23	5.0
Diving	22	19	25	6.0
Taekwondo	23	21	26	5.0
Synchronized Swimming	23	22	26	4.0
Boxing	23	21	25	4.0
Football	23	22	27	5.0
Gymnastics	23	20	27	7.0
Athletics	24	22	27	5.0
Cycling	24	21	27	6.0
Table Tennis	24	22	27	5.0
Rowing	25	22	28	6.0
Weightlifting	25	22	28	6.0
Hockey	25	23	28	5.0
Judo	25	23	27	4.0
Volleyball	25	23	28	5.0
Basketball	25	22	28	6.0
Water Polo	25	23	29	6.0
Badminton	25	23	27	4.2
Tennis	26	23	30	7.0
Softball	26	24	29	5.0
Wrestling	26	23	28	5.0
Handball	26	23	29	6.0
Canoeing	26	23	29	6.0
Baseball	26	23	29	6.0
Modern Pentathlon	27	25	29	4.0
Archery	27	22	38	16.0
Fencing	28	24	32	8.0
Sailing	30	26	36	10.0
Shooting	32	27	39	12.0
Equestrianism	34	29	41	12.0

Early-peak / tight-window sports (e.g. Rhythmic Gymnastics, median 18) reward physical peak; late-peak sports (Shooting 32, Equestrianism 34) reward accumulated experience — development pipelines should be sport-specific.

Appendix E — New LA28 Sports (scope addendum)

Five sports are added for LA28. **Four — cricket, flag football, lacrosse, squash — are absent from the historical data**, so they sit outside the quantitative zone model; the estimates below are forward-looking, anchored to comparable existing sports (Rio 2016 athlete counts). **Baseball and softball** were Olympic sports through Beijing 2008 and do appear in the data, so their figures are observed historical counts, not estimates.

New sport	LA28 venue / zone (per LA28 plan)	Comparable sport	Est. athlete load
Cricket (T20)	Fairgrounds — Pomona	Field Hockey (390)	≈ 390 (mid-size team sport)
Baseball	Stadium venue (returning sport)	Observed, Beijing 2008	191 (historical)
Softball	Stadium venue (returning sport)	Observed, Beijing 2008	120 (historical)
Lacrosse (Sixes)	Field venue — Carson cluster	Rugby Sevens (299)	≈ 299 (small-side team sport)
Flag Football	Stadium venue	Rugby Sevens / Basketball	≈ 281 (small-roster teams)
Squash	Temporary glass court	Table Tennis / Badminton (172)	≈ 172 (individual racket)

How this affects the core analysis. The added sports raise total athlete load but are spread across outlying / new zones (Pomona, stadium and field venues) rather than the three high-load zones, so they do not change the central finding: load remains concentrated in Downtown/Exposition Park, Long Beach, and Carson. Venue placements follow LA28 announcements and are noted for planning context only; they are outside our quantitative model.

Appendix F — Data Sources & Reproducibility

Dataset	Detail
Olympic athlete events	athlete_events.csv — 271,116 athlete-event rows, 1896–2016 (Kaggle / Sports Reference)
NOC regions	noc_regions.csv — National Olympic Committee to country mapping
World Bank — GDP per capita	Indicator NY.GDP.PCAP.CD (current US$), 2016
World Bank — Population	Indicator SP.POP.TOTL (total population), 2016
LA28 venue plan	Official LA28 venue-to-sport assignments (March 2026 map)

Tools & libraries

Python · pandas · NumPy · scikit-learn (PoissonRegressor, KMeans, KFold cross-validation) · SciPy · Matplotlib · Jupyter. Models: Poisson regression (NOC medal model) and KMeans clustering (country strategy).

Reproducibility

Every figure in this supplement and in the presentation traces to the notebooks in our GitHub repository: LA28_zone_load_analysis (Appendix A), gdpRegression (Appendix B), olympics_country_specialization_clustering (Appendix C), and olympics_analysis (Appendix D). A single script (build_supplement_data.py) regenerates every number in this document from the raw datasets, so SportsFanatics can audit or extend the analysis.